De La Salle University-Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSPC429
COURSE TITLE	**:**	Operating System
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
MODALITY	**:**	Traditional Blended Learning Model
PRE-REQUISITES	**:**	S-ITCS14LA
CO-REQUISITES	**:**	None
PROFESSOR	**:**	SHERYL D. KAMANTIGUE
CONSULTATION HOUR	**:**	Monday 9-10, Tuesday 8-10, COS100C Thursday 8-10, Friday 10-12 - Online
CONTACT INFORMATION	**:**	sdkamantigue@dlsud.edu.ph / MS Teams

Vision

A leading Catholic University that inspires excellence and drives innovation towards a just, peaceful and sustainable society.

Mission

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research and extension programs.

COURSE DESCRIPTION:

This course introduces a clear description of the concepts that underlie operating systems which focus on basic concepts, architectures, components, and theories that serve as bases for the design of classical and modern operating systems.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for the safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. understand operating system concepts by identifying how a computer system is organized;
CLO 2. explain how operating system accomplishes its tasks in different computing environments;
CLO 3. evaluate different aspects of operating systems, the various services the operating system provides and describe how operating systems are created;
CLO 4. analyze all vital concepts about processes;
CLO 5. examine the different scheduling algorithms utilized by different operating systems; and
CLO 6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Understand operating system concepts by identifying how a computer system is organized.	• TLO 1. Describe the general organization of a computer system and the role of interrupts. • TLO 2. Describe the components in a modern, multiprocessor computer system. • TLO 3. Illustrate the transition from user mode to kernel mode.
CLO 2. Explain how operating system accomplishes its tasks in different computing environments.	• TLO 4. Discuss how operating systems are used in various computing environments. • TLO 5. Provide examples of free and open-source operating systems.
CLO 3. Evaluate different aspects of operating systems, the various services the operating system provides and describe how operating systems are created.	• TLO 6. Identify services provided by an operating system. • TLO 7. Illustrate how system calls are used to provide operating system services. • TLO 8. Compare and contrast monolithic, layered, microkernel, modular, and hybrid strategies for designing operating systems. TLO 9. Illustrate the process for booting an operating system.
CLO 6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week/Date**	**Teaching-Learning Activities / Assessment Strategy** **Online Module Activities**	**Technology Enabler**	**Asynchronous**	**Onsite**	**Allotted Hours**
Module 1 TLO 1 TLO 2 TLO 3 CLO 1 CLO 6	Week 1 (Jan 20-25)	Gospel Reading and Reflection Syllabus discussion Presentation of course introduction Class Participation: Goal after graduation Discussion of Module 1 – Introduction Part 1 Enabling Assessment 1: Seatwork on What is the role of OS in multitasking? What are the real-life examples of multitasking? (20 points)	Schoolbook MS Teams PowerPoint		✓	2
Module 1 TLO 1 TLO 2 TLO 3 CLO 1 CLO 6	Week 2 (Jan 27-31; Feb 1-)	Gospel Reading and Reflection Continuation of Discussion of Module 1 – Introduction Part 1 Enabling Assessment 2: Onsite Short Quiz 1 on Module 1- Introduction (20 items)	MS Teams PowerPoint Schoolbook		✓	2
Module 2 TLO 4 TLO 5 CLO 2 CLO 6	Week 3 (Feb 3-8)	Gospel Reading and Reflection Discussion of Module 2 – Introduction Part 2 Enabling Assessment 3: Seatwork on Storage Management (20 items)	MS Teams PowerPoint		✓	2
Module 2 TLO 4 TLO 5 CLO 2 CLO 6	Week 4 (Feb 10-15)	Gospel Reading and Reflection Self-paced Reading of Module 2 – Introduction Part 2 Class Participation: Research on Virtualization	Schoolbook PowerPoint	✓		2
Module 2 TLO 4 TLO 5 CLO 2 CLO 6	Week 5 (Feb 17-22)	Gospel Reading and Reflection Finish Discussion of Module 2 – Introduction Part 2 Enabling Assessment 4: Onsite Short Quiz 2 on Module 2 (20 items)	Schoolbook MS Teams PowerPoint		✓	2

Module 2 TLO 4 TLO 5 CLO 2 CLO 6	Week 6 (Feb 24-29; March 1)	Gospel Reading and Reflection Self-paced reading of Module 3 – OS Services LA SALLIAN MISSION FESTIVAL	Schoolbook MS Teams PowerPoint	✓		2
Module 3 TLO 6 TLO 7 TLO 8 TLO 9 CLO 3 CLO 6	Week 7 (March 3-8)	Gospel Reading and Reflection Enabling Assessment 5: Long Quiz on Module 1 and Module 2	Pen and Paper Test		✓	2
Module 3 TLO 6 TLO 7 TLO 8 TLO 9 CLO 3 CLO 6	Week 8 (March 10-15)	Gospel Reading and Reflection Continuation of Self-paced Reading of Module 3 – OS Services	MS Teams Schoolbook PowerPoint	✓		2
	Week 9 (March 17-22)	MIDTERM EXAMINATION WEEK COVERING MODULES 1-3			✓	2
			TOTAL	***6***	***12***	***18***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 4. Analyze all vital concepts about processes.	• TLO 10. Identify the separate components of a process. • TLO 11. Illustrate how they are represented and scheduled in an operating system. • TLO 12. Describe how processes are created and terminated in an operating system, including developing programs using the appropriate system calls that perform these operations. • TLO 13. Describe client-server communication using sockets and remote procedure calls.
CLO 5. Examine the different scheduling algorithms utilized by different operating systems.	• TLO 14. Describe various CPU scheduling algorithms. • TLO 15. Assess CPU scheduling algorithms based on scheduling criteria. TLO 16. Apply modeling and simulations to evaluate CPU scheduling algorithms.
CLO 6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching-Learning Activities / Assessment Strategy** **Module Activities**	**Technology Enabler**	**Asynchronous**	**Onsite**	**Allotted Hours**
Module 4 CLO 4 TLO 10 TLO 11 TLO 12 TLO 13 CLO 6	Week 10 (March 24-29)	Gospel Reading and Reflection Self-paced reading of Module 4 – Processes SUSTAINABILITY WEEK	Schoolbook MS Teams PowerPoint	✓		2
	Week 11 (March 31-April 1-5)	Gospel Reading and Reflection Discussion of Module 5 – CPU Scheduling Discuss FCFS, SJF and SRTF			✓	2

		Enabling Assessment: Assignment on FCFS, SJF and SRTF				
Module 5 CLO 5 TLO 14 TLO 15 TLO 16 CLO 5 CLO 6	Week 12 (April 7-12)	Gospel Reading and Reflection Continuation of Discussion of Module 5 Discuss Priority Scheduling, Round Robin	MS Teams PowerPoint		✓	2
Module 5 CLO 5 TLO 14 TLO 15 TLO 16 CLO 5 CLO 6	Week 13 (April 14-19)	Gospel Reading and Reflection Self-paced reading on MLQ, MLFQ INDEPENDENT LEARNING WEEK HOLY WEEK	Schoolbook PowerPoint	✓		2
Module 5 CLO 5 TLO 14 TLO 15 TLO 16 CLO 5 CLO 6	Week 14 (April 21-26)	Gospel Reading and Reflection Enabling Assessment: Online Assignment regarding MLQ, MLFQ RESEARCH WEEK	Schoolbook PowerPoint	✓		2
Module 5 CLO 5 TLO 14 TLO 15 TLO 16 CLO 5 CLO 6	Week 15 (April 28-30; May 1-3)	Gospel Reading and Reflection Discuss MLQ, MLFQ Enabling Assessment: Seatwork on the last algorithms discussed	Schoolbook PowerPoint		✓	2
	Week 16 (May 5-10)	FINAL EXAMINATION WEEK COVERING MODULE 5			✓	2
			TOTAL	***6.0***	***8.0***	***14.0***

GRADING SYSTEM

Midterm

Class Participation/Attendance	-	20%
Enabling Assessments	-	50%
Summative Assessments	-	30%
Total		100%

Finals

Class Participation/Attendance	-	20%
Enabling Assessments	-	50%
Summative Assessments	-	30%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be

explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Online References:

Accession number or e-provider	**Reference Material**
e-Book	Silberschatz A, Galvin PB, Gagne G. *Silberschatz's operating system concepts* [Internet]. Wiley Global Education; 2020
Online	Networking Academy with Cisco. *Netacadcom* [Internet]. 2025 [accessed 2025 Jan 16]. Available from: https://lms.netacad.com/course/view.php?id=1816323

On-Site References

Provider	**References**
AEA Lib	Padallan J. *Introductory guide to operating systems*. Arcler Press; 2023.
AEA Lib	Persson J-B. *Linux system programming techniques*. Packt Publishing Ltd; 2021.

Journal References

Provider	**References**
Journal	Poplin A. Urban operating systems: producing the computational city. *J Urban Technol* [Internet]. 2022;29(4):130–132. Available from: https://doi.org/10.1080/10630732.2022.2130614
	Staso DD, Christiansen L, Kleiman F, Janssen M. Articulating social issues with open data: exploring a game jam approach. *[Journal Name]* [Internet]. 2024 Oct 11;[volume if available]([issue if available]):26–32. Available from: https://doi.org/10.1145/3697789.3697798

Prepared: **Computer Science Department**
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